

Heptadecagon

Gauss's 17-gon

Only a compass and a straightedge. Every point is marked on the front.

1. Quarter an angle

Bisect OB to find its middle, H. Bisect OH to find I, a quarter of the way up. Bisect angle OIA, then bisect the half next to IO. Where the line meets line OA: E.

2. 45 degrees, then a circle

At I, make a right angle to IE. Mark two points on line IE, one on each side of I, the same distance away. From each, draw a wider arc. Join I to where the arcs cross. Bisect the right angle on O's side of IE to get 45 degrees. Where it meets line OA: F. Find the middle of AF: equal arcs from A and from F, then join their crossings. The middle is M. Compass on M, opened to A: draw the circle. It meets OB at K.

3. One more circle

Compass on E, opened to K: draw the circle. It meets line OA at N3 and N5. Raise a perpendicular at each, up to the big circle: P3 and P5. N3 and M are only 0.74 mm apart, so their pits look like one.

P3 is exactly $\frac{3}{17}$ of the way round from A. P5 is exactly $\frac{5}{17}$.

4. Walk it round

Open the compass from P3 to P5 and step it from A toward P3: you land $\frac{2}{17}$ of the way round. From there to P3 is one side, exactly $\frac{1}{17}$. Walk that width round 17 times, checking each mark against its tick, and join the marks.

Inspired by Numberphile:
Heptadecagon (17-gon)
Scan for the video.

